# Linear Regression

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- Supervised. ↳ Ordinary Least Square (OLS)

Types:
- Simple LR
- Multiple LR
- Polynomial.

# Simple LR:

1 input column & 1 output column.

| cgpa | package |
|---|---|
| 6.6 | 5.2 |
| 7 | 6 |
| 9 | 15 |

# Multipe LR: multiple input

| State | cgpa | iq | Package |
|---|---|---|---|
| | | | |

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# Simple Linear Regression:

$$y = mx + b$$

General: plt.plot(X_test, y_pred)

Training : plt.plot(x_train, y_pred(X_train)).

## Solution:

- closed form solution. [Linear Regression].
  do exact to solve eq$^n$
  quadratic $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.
- Non-closed form solution.
  approximation: [integrate, differentiate].
  → Gradient descent.

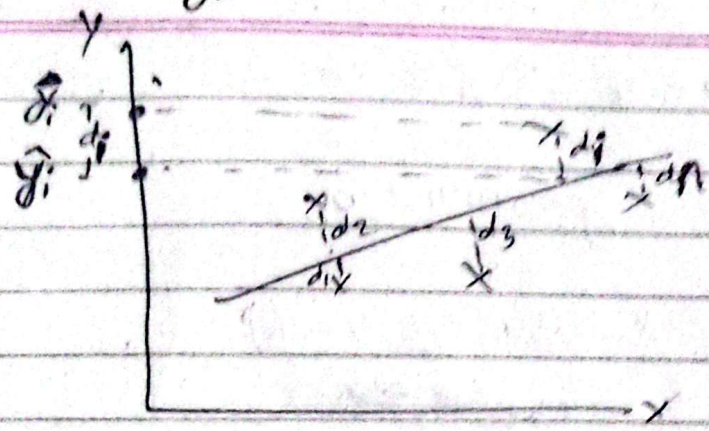
Using closed form will be difficult in higher dimension. so, non-close is also req$^d$.

$$\left\{ \begin{aligned} b &= \bar{y} - m\bar{x} \\ m &= \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \end{aligned} \right\}$$

$\hat{y}_i \rightarrow$ prediction data.

$y_i \rightarrow$ real time data






$$E = d_1 + d_2 + d_3 + \cdots + d_n.$$

As distance will be -ve as well as +ve which may cancel each other.

Hence: use

$$E = d_1^2 + d_2^2 + d_3^2 + \cdots + d_n^2$$

$$\left\{ \underset{\text{or } J}{E} = \sum_{i=1}^{n} d_i^2 \right\} \text{ error } f^n / \text{ loss function.}$$

Why not $E = |d_1| + |d_2| + \ldots$

- not diff in 0
- we have to analyze all the outliers.

- we need values of m & b, which lower the value of loss function.

$$d_i = (y_i - \hat{y}_i)$$

$$\left\{ E = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \right\}.$$

average error

$$E = \sum_{i=1}^{n} \frac{(y_i - \hat{y}_i)^2}{n}.$$

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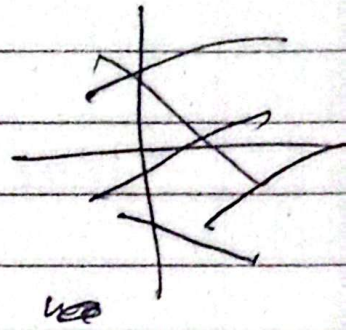
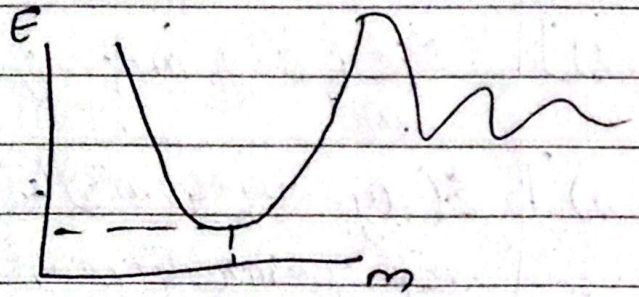
$\hat{y}_i = m x_i + b.$

$$E(m,b) = \sum_{i=1}^{n} (y_i - m x_i - b)^2$$



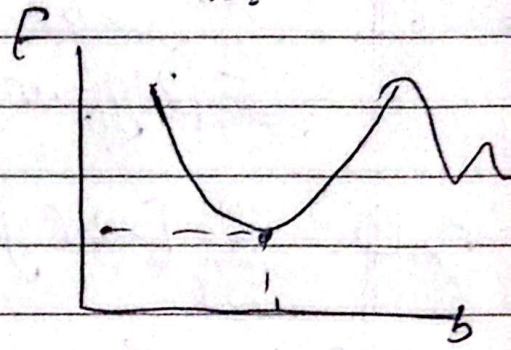
When $b = 0$. [passes through origin].

$$E(m) = \sum_{i=1}^{n} (y_i - m x_i)^2$$

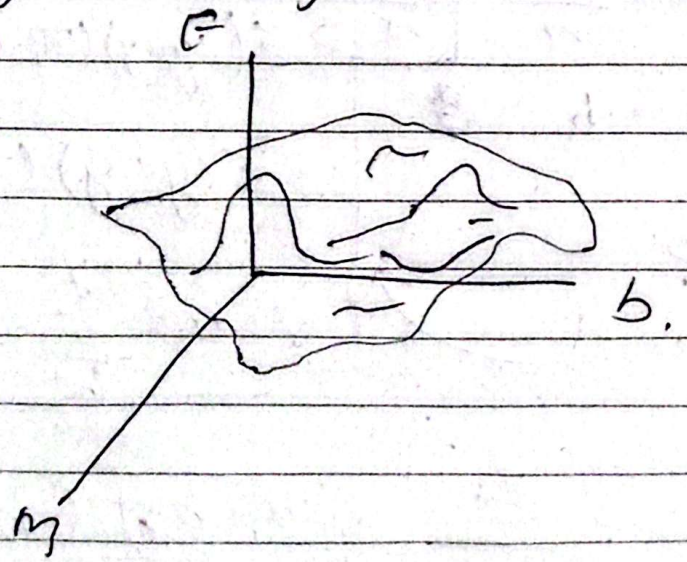



When $m = 1$

$$E(b) = \sum_{i=1}^{n} (y_i - x_i - b)^2$$




Taking both together.




Use maximal minima.

~~differentia~~

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for minimum slope = 0;
differentiate & do 0

Here; 2 ~~f~~ feature:

$$\frac{\partial(E)}{\partial b} = 0$$

$$\frac{\partial E}{\partial b} = \frac{\partial}{\partial b}\left(\sum_{i=1}^{n}(y_i - mx_i - b)^2\right)$$

$$= \sum_{i=1}^{n} \frac{\partial}{\partial b}(y_i - mx_i - b)^2$$

$$= \sum_{i=1}^{n}\left[2(y_i - mx_i - b)\cdot(-1)\right]$$

$$\Rightarrow 0 = \sum_{i=1}^{n}\left[-2(y_i - mx_i - b)\right]$$

$$\Rightarrow \frac{0}{n} = \sum\left(\frac{y_i}{n} - \frac{mx_i}{n} - \frac{b}{n}\right)$$

$$\Rightarrow 0 = \frac{\sum y_i}{n} - \frac{\sum mx_i}{n} - \boxed{\frac{\sum b}{n}}$$

$$\Rightarrow 0 = \bar{y} - m\bar{x} - b$$

$$\bar{y} = m\bar{x} + b$$

$$\frac{\partial(E)}{\partial m} = 0$$

$$\Rightarrow \frac{\partial}{\partial m}\left(\sum_{i=1}^{n}(y_i - mx_i - b)^2\right) = 0.$$

$$\Rightarrow \cancel{\sum\left[2(y_i - mx_i - b)(-x_i)\right] = 0}$$

$$\Rightarrow \frac{\partial}{\partial m}\left(\sum(y_i - mx_i - \bar{y} + m\bar{x})^2\right) = 0$$

$$\Rightarrow \sum\left(-2(y_i - mx_i - \bar{y} + m\bar{x})(-\bar{x} + x_i)\right) = 0$$

$$\Rightarrow \sum\left\{(y_i - mx_i - \bar{y} + m\bar{x})(x_i - \bar{x})\right\} = 0$$

$$\Rightarrow \sum\left\{(y_i - \bar{y}) - m(x_i - \bar{x}))(x_i - \bar{x})\right\} =$$

$$\Rightarrow \sum\left((y_i - \bar{y})(x_i - \bar{x}) - m(x_i - \bar{x})^2\right) = 0$$

$$\Rightarrow \sum(y_i - \bar{y})(x_i - \bar{x}) = \sum m(x_i - \bar{x}$$

$$\therefore m = \sum_{i=1}^{y}\frac{(y_i - \bar{y})(x_i - \bar{x})}{(x_i - \bar{x})^2}$$

$$\therefore m = \sum_{i=1}^{y}\frac{(x_i - \bar{x})(y_i - \bar{y})}{(x_i - \bar{x})^2}$$

$$y = \bar{y} - m\bar{x}$$

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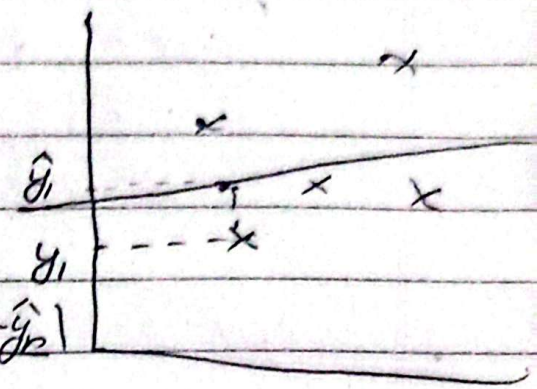
# Regression Metrics (Accuracy test).

## 1) Mean Absolute Error [MAE]

$d_1 = |y_1 - \hat{y}_1|$
$d_2 | y_2 - \hat{y}_2 |$




$$MAE = \frac{|y_1 - \hat{y}_1| + |y_2 - \hat{y}_2| - - - - + |y_n - \hat{y}_n|}{n}$$

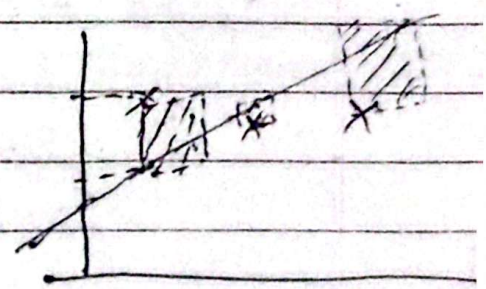
$$MAE = \sum_{i=1}^{n} \frac{|y_i - \hat{y}_i|}{n}$$

**Advantages:**
- Handel outliers

**Disadvantage:**
- Not diff at zero.

## 2) Mean Square Error [MSE].

$$MSE = \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n}$$



- ~~Diff~~

**Ad:**
Diff at all points

**Disadv:**
- Can't handle outliers.
- Unit
$m \longrightarrow m^2$

* calculate all error and check.

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3) Root Mean Square Error [RMSE].

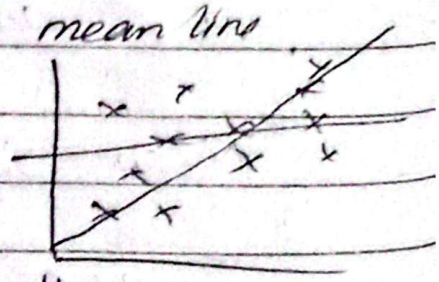
$$R_{mse} = \sqrt{msc}$$

$$= \sqrt{\frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n}}$$

benefit:
- like mse

demerit:
- not more robust to outliers.

4) R2 score
- Compare how much better is predict after regression as compared to mean line



$$R_2 = 1 - \frac{SSR}{SSM}$$

SSR = sum of sq in regression line
SSM = sum of sq in mean line

$$R_2 = 1 - \frac{\left[\sum_{i=1}^{n} (y_i - \hat{y}_i)^2\right]_{Reg}}{\left[\sum_{i=1}^{n} (y_i - \hat{y}_i)^2\right]_{m}}$$

- $R_2 = 0 \Rightarrow$ mean line = regression line. [worst case].
- $R_2 = 1 \Rightarrow SSR = 0$: regression line is perfect.

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- $R_2 = -1 \Rightarrow SSR > SSM$ [not working]. [use another way]

If $R_2 = 0.8$ means (lpa)
80% of variance in data can be explained by feature data (cgpa).

- Disadvantage:
  - Higher the features, higher will be the $R_2$ score (in cgpa & lpa, adding temp' column of that day, ~~remo~~ $R_2$ remains same or increases, but it needs to be decreased!

5) Adjusted $R_2$ Score.

$$R^2_{adj.} = 1 - \left[\frac{(1-R_2)(n-1)}{(n-1-k)}\right]$$

k = number of columns/features

Don't do if there is pattern betn temp' & lpa.